

Session 1 — Anticipated Participant Queries & Trainer Responses

Azure Core Architecture & Services

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For trainer use: these are questions participants commonly raise during or after this material. Each answer is written the way you should deliver it verbally — direct, correct, and free of hedging where the facts are settled.

Global Infrastructure

Q: If a Region Pair protects us from disaster, why do we need Availability Zones at all?

A: They protect against completely different scales of failure. A Region Pair protects you if an entire region — potentially many datacenters — becomes unavailable, which is rare but catastrophic. An Availability Zone protects you against something far more common: one datacenter losing power, cooling, or network connectivity while the rest of the region keeps running fine. Most production-grade designs use zone-redundancy as the default and treat cross-region failover as the extreme fallback.

Q: Can I choose which Availability Zone my VM lands in, or does Azure choose for me?

A: You can pin a resource to a specific zone (a "zonal" deployment) or let Azure spread it automatically across zones (a "zone-redundant" deployment). In today's lab we explicitly pinned the VM to Zone 1 using `--zone 1` so the concept is visible — in a production high-availability design, you'd typically deploy identical VM instances across two or three zones behind a load balancer, which we'll build in Session 2.

Q: What happens if we pick a region that isn't Availability-Zone enabled?

A: You simply won't have the option to deploy zone-redundant resources there — the portal and CLI will not offer zone parameters for that region. You'd rely on a Region Pair for resilience instead. This is exactly why we stressed "verify availability before designing" — it directly affects what resilience patterns are even possible.

Q: Is data ever allowed to leave the geography we chose?

A: Only if you explicitly configure it to — for example, choosing GRS replication to a paired region that happens to sit in a different geography, or manually copying data elsewhere. By default, Azure keeps data within the geography you deployed into.

ARM & Resource Organization

Q: Do I have to use ARM templates or Bicep, or can I just always use the Portal?

A: You can use the Portal for everything if you want — it's a completely legitimate way to work, especially while learning. The tradeoff is repeatability: a template lets you redeploy the exact same environment reliably, which matters once you're managing more than a handful of resources or need to recreate environments (Dev, Test, Prod) consistently.

Q: If I delete a Resource Group by mistake, is there any way to recover it?

A: Not reliably, which is exactly why we apply Resource Locks. Some individual resource types have soft-delete

or retention features (Storage Accounts and Key Vaults, for example), but there is no guaranteed "undo" for a deleted Resource Group as a whole. Locks are the real safeguard — that's the entire point of applying one in today's lab.

Q: What's the actual difference between a Management Group and a Subscription for organizing resources?

A: A Subscription is where billing and default quotas live — it's a hard boundary. A Management Group sits above subscriptions purely for governance — applying policy and access control across many subscriptions at once. You can't put resources directly inside a Management Group; resources always live inside a Resource Group, inside a Subscription.

Q: Can one subscription belong to more than one Management Group?

A: No — a subscription belongs to exactly one management group at a time, though it can be moved between them as the organization's structure changes.

Q: We only have one subscription today — is any of the Management Group content actually relevant to us?

A: Yes, because this structure is designed to scale. Even a single subscription still sits under the tenant's root management group by default, and understanding the hierarchy now means you won't have to relearn it when BFL eventually splits workloads across multiple subscriptions — which most growing organizations do.

Azure Portal

Q: Is Cloud Shell free to use?

A: The Cloud Shell session itself has no separate charge, but it does require a small amount of Azure Files storage to persist your session, which carries a minimal storage cost. Any resources you create using Cloud Shell are billed normally like any other Azure resource.

Q: Does everyone on the team see the same Portal dashboard?

A: No — Dashboards and Favorites are personalized per user by default. You can optionally share a dashboard with others if you want a common view for the team, but each person's default view is their own.

Compute

Q: How do I know if I should use App Service or a plain Virtual Machine for a web application?

A: Default to App Service unless you have a specific reason not to — it removes OS patching and infrastructure management entirely. Reach for a Virtual Machine when you need something App Service can't give you: a custom OS configuration, specific licensed software that requires full server access, or an existing application that wasn't built for a PaaS environment.

Q: Is Azure Functions only for tiny scripts, or can it run something substantial?

A: Functions can run fairly substantial logic, but it's designed around short execution windows and event triggers rather than long-running processes. If your workload needs to run continuously or for extended periods regardless of events, a VM, App Service, or a container-based service is the better fit.

Q: What's the real difference between AKS and Container Instances if they both run containers?

A: Scale and orchestration. Container Instances runs one or a few containers with no orchestration layer — good for a quick, simple job. AKS is a full Kubernetes cluster, built for running many interdependent

containerized services with scaling, self-healing, and networking between them. If you're only running one container, AKS is almost always overkill.

Storage

Q: If ZRS already protects against a datacenter failure, why would anyone still choose GRS?

A: ZRS only protects within one region — if the entire region became unavailable, ZRS alone wouldn't help. GRS protects against that scenario by asynchronously replicating to a paired region. The tradeoff is cost and replication lag: GRS costs more and isn't synchronous, so there's a small window where the secondary copy may lag behind. Choose based on how severe an outage you need to survive.

Q: Can I change a storage account's redundancy option after it's created?

A: In most cases yes, you can change the redundancy tier after creation, though some transitions (particularly moving into or out of GRS/ZRS combinations) can take time to fully replicate and may have specific conversion requirements. Always check the current conversion rules on Microsoft Learn before promising a specific migration path to stakeholders.

Q: Do Managed Disks use the same redundancy options (LRS/ZRS/GRS) as Blob Storage?

A: Managed Disks have their own redundancy behavior tied to the disk type and zone configuration rather than the exact same LRS/ZRS/GRS menu as Storage Accounts — always check the current Managed Disk documentation for what's supported for the specific disk tier you're using.

Reference: Microsoft Learn (learn.microsoft.com/azure). Service behavior, limits, and conversion rules evolve — verify anything time-sensitive against the live documentation before presenting it as current fact.